Class Diagram - System Structure (AirPrice)

ScraperApp
(tk.Tk)

+bool scheduler_running
+Event stop_event
+WebDriver current_driver
+dict frames

+show_frame()
+log_ok() / log_err() / log_warn() / log_info()
+request_stop()
+set_scheduler_status()

1 instance 1

StartPage
(ttk.Frame)

+update_scheduler_badge()

1 instance 1

ExperimentoPage
(ttk.Frame)

+rota_var, +plataforma_var
+data_voo_var, +repeticoes_var

+_iniciar_coleta()

1 instance 1

AgendadorPage
(ttk.Frame)

+janelas_dias, +intervalo_min

+_iniciar()
+set_next_run()
+update_status_badge()

1 instance 1

HistoricoPage
(ttk.Frame)

+listbox

+_atualizar()
+_enviar_selecionado()

1 instance 1

ConfigPage
(ttk.Frame)

+cred_var, +plan_var, +aba_var
+proxy_enabled, +proxy_list

+_salvar()
+_selecionar_cred()

ConsoleController
(coleta_tcc.py)

+log_ok() / log_err()
+log_warn() / log_info()
+main()

triggers (start_thread)

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triggers (resend)

calls executar_rotas_sequencial()
(same logging interface as ScraperApp)

Orquestrador de Coleta
(scraper_passagens.py)

+ROTAS_DISPONIVEIS
+COLUNAS_DATASET
+stop_event

+criar_driver()
+executar_experimento_thread()
+executar_plataformas_sequencial()
+executar_rotas_sequencial()
+executar_scheduler_thread()
+salvar_excel()
+enviar_google_sheets()
+salvar_e_enviar()

uses (scrape_latam)

uses

LatamScraper
(latam_scraper.py)

+URL_TEMPLATE
+MAX_TENTATIVAS_CARREGAMENTO

+scrape_latam()
-_detectar_bloqueio()
-_fechar_banner_cookies()
-_aguardar_cards() / _buscar_cards()
-_extrair_menor_preco_com_fallback()

uses

uses

ScrapersUtils
(scrapers_utils.py)

+ColetaCancelada (Exception)

+_montar_registro()
+extrair_menor_preco()
+hash_gz_file()
+destruir_html_log()
+sleep_interrompivel()
+_gerar_id() / _classificar_janela()